MD ZOBAYER HOSEN



Full-Stack Developer | Next.js , NestJS & TypeScript | AI/ML Enthusiast

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github.com/zobayer-hosen | Portfolio

Career Objective

Disciplined final-semester CSE student and software developer with strong analytical, problem-solving, and programming skills. Seeking opportunities in software development and artificial intelligence to build scalable applications, develop intelligent solutions, and solve real-world problems through innovative technologies.

Professional Summary

- **Full-Stack Developer** at **Bee Smart IoT Solutions**, building web applications with Next.js, NestJS, TypeScript, and PostgreSQL.
- Strong interest in **backend fundamentals**: API design, database architecture, authentication, concurrency, and system design.
- Exploring **Go (Golang), Docker, and CI/CD** to build secure, reliable, and scalable backend systems.
- **AI/ML enthusiast** exploring Explainable AI, AI agents, and agentic workflows.

Key Skills

Programming Languages		JavaScript, TypeScript, Python, Go (Golang), C++, C#
Frontend Development		React.js, Next.js, HTML5, CSS3
Backend Development		Node.js, NestJS, Go, REST APIs, PostgreSQL, SQL, Low-Level Design (LLD)
DevOps & Tools		Git, GitHub, Docker, CI/CD Pipelines, Swagger, Linux
AI Engineering		LLM APIs, LangChain, Prompt Engineering, RAG, AI Agents, Agentic Workflows
AI/ML		Machine Learning, Explainable AI (XAI), Data Analysis

Professional Experience

Bee Smart IoT Solutions – *Full-Stack Developer Intern* | Dhaka, Bangladesh **Aug 2026 – Present**

- Developed the official websites of **Bee Smart IoT Solutions** and **Nexoit Solution** using **Next.js, NestJS, TypeORM, and PostgreSQL**.
- Build **RESTful APIs** and responsive user interfaces for **IoT-based monitoring and control** solutions.
- Collaborate with the developer team on the company repository using **Git, code reviews, and Agile workflows**.

Head Blocks – *AI/ML Intern* | Dhaka, Bangladesh **Mar 2024 – May 2024**

- Conducted research on **machine learning** models, including data preprocessing, model training, and evaluation using **Python**.
- Explored **Explainable ML** techniques to interpret model predictions and improve transparency.

Projects

Dhaka Tesla Pool – Ride-Pooling Platform Next.js, NestJS, PostgreSQL, TypeORM, Docker
`https://github.com/zobayer-hosen/dhaka-tesla-pool` | Live Demo

- Built a ride-pooling MVP where passengers heading the same way share a 3-seat car, with separate passenger and driver flows and a strict ride lifecycle (requested to completed).
- Prevented overbooking under simultaneous requests using a single atomic SQL update backed by a CHECK constraint, verified with a concurrent race test.
- Shipped with 91 passing tests and GitHub Actions CI; deployed on Vercel, Render, and Neon.

EchoGPT Backend – Multi-AI Chrome Extension API NestJS, PostgreSQL, TypeORM, TypeScript
`https://github.com/zobayer-hosen/echogpt-backend` | Live Demo

- Developed a REST API that lets users chat with OpenAI, Anthropic, and Gemini models and run AI-assisted searches through one interface, with streaming responses over Server-Sent Events.

- Implemented rotating refresh tokens with database-backed sessions, Free/Premium plans with atomically enforced daily quotas, and AES-256-GCM encryption of provider keys.
- Designed a pluggable provider-adapter architecture with admin APIs, Swagger docs, and end-to-end tests.

Hospital Management System Next.js, React.js, NestJS, TypeORM, PostgreSQL
https://github.com/zobayer-hosen/hospital-management-system

- Developed a full-stack hospital management platform with separate dashboards for patients, doctors, receptionists, and system administrators.
- Implemented JWT-based authentication, protected API routes, and fine-grained Role-Based Access Control (RBAC) with scalable NestJS REST endpoints.

Research Experience

Phenology-Aligned Multi-Stress Crop Yield Prediction with Stage-Specific Climate Sensitivity *Ongoing*

- Designing **explainable machine learning** models for crop yield estimation by aligning localized **phenological stages** with environmental and climate stress indicators.
- Applying explainability techniques to quantify how climate stress at each growth stage affects final yield.

Detection of APT Command-and-Control (C2) Communication Using Deep Packet Inspection *Ongoing*

- Investigating **Deep Packet Inspection (DPI)** frameworks and network traffic heuristics to detect stealthy **Advanced Persistent Threat (APT)** command-and-control communication.

Academic Qualifications

B.Sc. in Computer Science & Engineering **Expected: Dec 2026**
American International University–Bangladesh (AIUB), Dhaka CGPA: 3.41 / 4.00

Higher Secondary Certificate (H.S.C.), Science **2022**
Chattogram Bandar College, Chattogram GPA: 5.00 / 5.00

Secondary School Certificate (S.S.C.), Science **2020**
Julon Pole BM High School, Chattogram GPA: 4.67 / 5.00

Achievements & Certifications

- **2nd Runner-Up, AIUB FST Apps Showcasing:** Recognized for developing the *AIUB-Chatbot* project.
- **Internship Offer, Head Blocks:** Selected for an AI/ML internship based on performance in a competitive selection process.
- **Certifications & Verified Credentials:** View Credentials Folder

Language Proficiency

English: Professional Working Proficiency | **Bengali:** Native

References

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